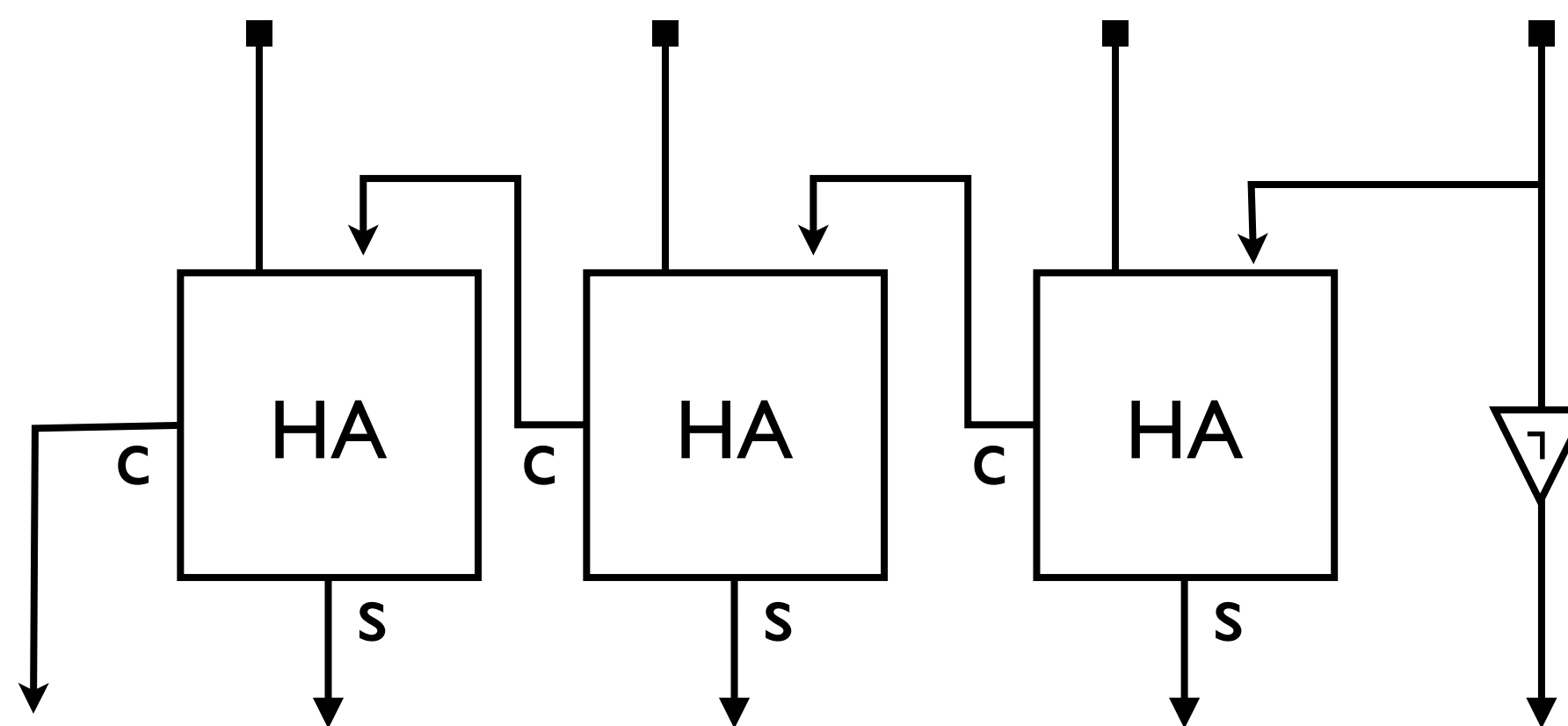


16. Lessons and Limits of Combinatorial Computation

5 Fundamental Lessons of CC computation



+1[0-15] B combinatorial circuit

Lesson 1: The physical realization of a cognition.

+1B shows us definitively that an entirely physical system can perform cognitive operations, via the method of symbolic simulation.

Lesson 2: Information processing as functional description.

Hammers have the function of pounding nails, and hearts have the function of pumping blood. +1B has an *information processing* function, which goes beyond the way it manipulates the physical world. “It computes +1” is the informational-level description of the machine. Compare +1B and +1T: different physical functions, same information processing function.

Lesson 3: Computing a function vs implementing a function.

All things *implement* the laws of physics, but only some things *compute* the laws of physics. To compute is to *represent* a function, but not to actually conform with it. +1B can compute the laws of *adding-apples-to-a-basket*, but it does not *implement* these laws. (By the way, do the planets compute their own orbit? When humans walk, do they compute or implement their trajectory?)

Lesson 4: Computation does not require an explicitly represented algorithm.

+1B computes a function not by “following” a set of rules, which would require an inner-agent to do the following, but simply by *embodying* a set of rules in its functional architecture. To compute a function is to

Lesson 5: Competence vs performance.

+1B shows how a physical system can have certain cognitive *capacities*— i.e. the +1B machine has the capacity to compute +1, for a range of arguments, even if it is never actually used for this. It also illustrates the contrast between the capacities you have *competence* for, and your actual performance. Because of entropy, friction, and the rest, how a machine actually performs can pull apart from its competence.

1. **Multi-functionality.** Natural cognitive systems compute many different functions. Our +1B machine computes only one.
2. **Learning.** Natural cognitive systems change their behavior in response to experience. The behavior of our +1B machine is fixed for eternity.
3. **Non-arithmetical functions.** Natural cognitive systems perform non-arithmetical functions, including decision making, action-planning, prediction, navigation. What can a CC do beyond +1B?
4. **Logical and probabilistic inference.** Among non-arithmetical functions some employ language-like reasoning, especially logic and inference. +1B is stuck in a world of integers.
5. **Perception and action.** Natural cognitive systems exist within an environment. They gather information from that environment (perception), and they change that environment (action). +1B has no obvious causal connection to the natural order.
6. **Biological basis.** Brains are built from neurons, not logic gates; neurons are put together to form neural nets, not combinatorial circuits. So +1B fails to capture the physical and functional basis of natural computation.
7. **Emotion and consciousness.** +1B is incapable of emotion or consciousness. (Right?) It is not clear, from the current standpoint, what would even be required to achieve this.
8. **Original functionality.** Natural cognitive systems have the biological function of performing certain information-processing tasks, as a result of natural selection. The +1B machine's only function was stipulated in a problem set.